

GreenSort — Smart Waste Management

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Abstract

GreenSort is a smart waste management system using AI and IoT.

YOLO and Random Forest models were used for waste classification.

The system categorized waste into four major categories with approximately 75% accuracy.

A Raspberry Pi controlled servo motors for automated waste sorting.

The project resulted in a research paper publication at SustainX 2026.

Technologies Used

Technology Stack
YOLO, OpenCV, Arduino, Raspberry Pi

Future Scope

This project can be extended with advanced AI models, cloud deployment, real-time scalability, and improved datasets for better accuracy and performance.

Conclusion

The project demonstrates practical implementation of AI, machine learning, or software technologies to solve real-world problems effectively.